PALATIAL POOL SERVICES

ESP32 Diagnostic Tools

Precision IoT Sensor Arrays for Complex Diagnostics

Our custom-engineered IoT diagnostic arrays deploy temporarily to investigate complex pool issues that standard equipment cannot detect. These sensor suites monitor water chemistry micro-fluctuations, electrical current draw, and fluid dynamics at high frequency — identifying intermittent faults, stray voltage, and chemical imbalances with laboratory-grade precision.

Key Benefits

- Detects intermittent faults invisible to standard equipment
- Identifies dangerous stray voltage in pool water
- Laboratory-grade chemical analysis without lab delays
- Non-invasive installation — no plumbing or electrical modifications
- 48-hour battery backup ensures uninterrupted monitoring

Technical Specifications

Processor	Dual-core ESP32 at 240 MHz with 520KB SRAM
Sensors	pH, ORP, TDS, current transformers, pressure transducers, flow meters
Sampling Rate	10Hz telemetry, aggregated into 1-second payloads
Connectivity	Wi-Fi 802.11 b/g/n, Bluetooth 4.2, LTE-M/NB-IoT fallback
Battery	18650 lithium-ion, up to 48 hours backup

Deployment	IP67-rated Pelican case, 3-7 day monitoring periods

Experience the Difference

Contact our team for a personalised demonstration of this technology.

Call Us: 0452 157 694 | service@palatialpools.com.au

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